

BONE STORIES - EDUCATIONAL KIT

Activity 4

One bone, many lives

A method workshop to describe a bone or an object, propose several hypotheses and specify how to verify them without transforming a possibility into a certainty.

AUDIENCE**9-12 years old****DURATION****45-75 mins****CONTENT****Guide, sheet, maps, answer key****LENGTH****16 pages**

In this kit

1. Facilitator guide
2. Learner sheet
3. Object cards A to E to complete
4. Method correction
5. Sources and limits

Careful prototype - method supports; objects A to E to be completed after validation.

Recommended printing: guide and correction for adults; student by student record; cards on thicker paper if they are cut out. The institution's validated objects can be observed directly or added to their intended locations.

Version: 1.0 pilot - July 20, 2026.

BONE STORIES - ACTIVITY 4

Facilitator guide

One bone, many lives

Guide to conducting an investigation into the possible uses of an object without announcing a definitive function.

TYPE
Facilitator guide

AUDIENCE
Supervising adults

DURATION
45-75 mins

VERSION
Pilot 1.0

Cautious prototype - objects A to E and their instructions remain to be integrated.

1. In brief

The students describe an object without attributing a use to it, formulate at least two hypotheses, choose the clues that support them then list the information necessary to decide between them. Cards A to E are templates: the activity becomes fully usable after adding a validated corpus.

2. Objectives

- Distinguish observation, hypothesis, interpretation and verification.
- Describe material, shape, dimensions and visible traces.
- Propose several possible uses from clues.
- Understand the role of the archaeological and documentary context.

3. Expected success

Successful

Two hypotheses are linked to clues and to means of verification.

In progress

The hypotheses are distinct, but the clues remain imprecise.

To resume

The function is announced as certain from the first observation.

4. Duration and material

Context	Duration	Material
Class	45-75 mins	Sheet, A-E cards, photographs or validated replicas.
Museum	30-60 mins	Object in display, reproduction or manipulatable object authorized.

5. Quick preparation

À PRÉPARER

Select one to five documented objects. Complete the cards without writing a function as a title. Keep context data hidden until the verification phase. Adapt the protocol to the rules of the institution.

À DIRE

“Start with what is visible. A hypothesis is a proposition to be tested, not a certain answer. »

6. Detailed procedure

Time	Adult action	Student action	Purpose
5 mins	Present the object without function.	Write a first impression.	Bring out initial ideas.
10 mins	Temporarily ban common words.	Describe material, shape, traces, dimensions.	Install observation.
15 mins	Ask for at least two hypotheses.	Link each to clues.	Open multiple interpretations.
10 mins	Distribute validated context data.	Confirm, qualify or dismiss leads.	Test assumptions.
10 mins	Conduct a collective conclusion.	Classify established, plausible and unknown.	Conclude with caution.

7. Questions and reminders

Observe

- What can you measure?
- What traces are visible?
- What does this view not show?

Check

- What context would decide between the hypotheses?
- Which reference object to compare to?
- What analysis would be useful?

8. Responses or expected elements

No A-E function is expected before integrating the corpus. A successful production contains an independent description, at least two hypotheses, the associated clues, an objection or limit and a means of verification: context of discovery, traces of use, typological comparison, analysis of material, measurement or textual source.

9. Differentiation

Need	Adaptation
Help	Work on a single object and provide vocabulary labels.
Lighten the writing	Classify "I observe/I suppose/I verify" cards.
Go deeper	Compare two records and discuss what changes with provenance.

10. Museum adaptation

Choose an object adapted to the institution and its actual collections. Observation can be done in the window; a photograph or replica is a possible variation but does not entirely replace the original. Manipulation is only considered if the object can be handled, if the museum rules authorize it and according to the conservation protocol.

11. Quick assessment

Highlight in a conclusion an observation, a hypothesis and a means of verification. Evaluate the ability to change your mind when the context brings a new clue.

12. Common mistakes

À NE PAS AFFIRMER

Do not assert that a knucklebone was necessarily used for playing, predicting or counting. Do not announce an undocumented museum validation. Do not present a modern digital adaptation as a historical reconstruction.

- Name the object before describing it.
- Keep only one hypothesis.
- Confusing resemblance and same usage.
- Ignore discovery context.

13. Sources and limits

The support transposes an archaeological investigation approach: observation, formulation of hypotheses, comparison with the context and comparison. The five Metropolitan Museum of Art images documented as provisional in the redesign are excluded from the PDFs. The coding game is a contemporary extension, not proof of ancient use.

14. Current status and remaining validations

À VÉRIFIER

Templates and method ready for adult testing. Before a student session, integrate documented A-E objects, check notices and rights, establish the information revealed at each stage and have the corpus validated by scientific and museum managers.

Print plan

Each student: student sheet.

Each group: one to five completed object cards.

Adult: guide, correction and complete instructions.

Cutting: single-sided cards, 120-160 g paper.

BONE STORIES - ACTIVITY 4

Learner sheet

One bone, many lives

TYPE

Learner sheet

AUDIENCE

9-12 years old

DURATION

45-60 mins

VERSION

Pilot 1.0

Cautious prototype - to be used with an object chosen and validated by the adult.

First name:

Date:

Subject:

Your mission

Observe an object without concluding too quickly, propose several possible stories and explain how to verify them.

Objective: learn to distinguish what you see, what you assume and what you can verify.

1. I observe

Write only what is visible or measurable. Does not yet indicate the use of the object.

Material / color	Shape / dimensions	Traces / reliefs	view missing

Word bank: curved - polished - perforated - broken - regular - asymmetrical - trace - surface - measurement.

2. I question

Choose two details that require explanation.

3. I propose several hypotheses

Hypothesis	clue who supports it	What weakens it	How to check
1			
2			

4. I discover the context

After the validation of the information is revealed, what changes in your hypotheses?

5. My cautious conclusion

Classify your ideas: what is established, what is plausible and what remains unknown.

Established	Plausible	Still unknown

À OBSERVER

A solid interpretation links multiple clues and remains editable if new data appears.

Traceability: see the activity page and the project's credits document.

BONE STORIES - ACTIVITY 4

Object cards to complete

One bone, many lives

Templates to be completed only with a corpus whose notices, relevance and rights have been validated.

TYPE**Object cards****AUDIENCE****Groups****USAGE****Cutting or projection****VERSION****Pilot 1.0**

Objects A to E to complete - no archaeological functions or images are provided in this version.

Instructions for use for adults

- 1 Replace the neutral zone with a validated photograph or present the object in a window.
- 2 Complete only the data visible at the start of the survey.
- 3 Keep provenance, dating, context and references for the verification phase.
- 4 Never include the supposed function in the title of the card.

Objects A and B

Object A

TO COMPLETE

Validated image or object to be integrated by the institution.

Visible material:

orientation:

Dimensions:

view:

Starting question:

Object B

TO COMPLETE

Validated image or object to be integrated by the institution.

Visible material:

orientation:

Dimensions:

view:

Starting question:

Objects C and D

Object C

TO COMPLETE

Validated image or object to be integrated by the institution.

Visible material:

orientation:

Starting question:

Dimensions:

view:

Object D

TO COMPLETE

Validated image or object to be integrated by the institution.

Visible material:

orientation:

Starting question:

Dimensions:

view:

Object E

Object E

TO COMPLETE

Validated image or object to be integrated by the institution.

Visible material:

orientation:

Dimensions:

view:

Starting question:

Context card to reveal

Origin:

Dating:

Analyzes/comparisons:

Rights and credit of the image:

Institution/inventory:

Context of discovery:

Scientific source:

BONE STORIES - ACTIVITY 4

Method correction

One bone, many lives

TYPE

Adult answer key

AUDIENCE

Teachers and museum educators

USAGE

Group discussion

VERSION

Pilot 1.0

Reasoning correction - no functions A to E are defined in this version.

Correction grid

Step	Expected	Acceptable example	To resume
Observation	Visible or measurable characteristics.	"Polished surface on one side".	"Game object".
Hypothesis	At least two distinct possibilities.	Wear linked to use or natural transformation.	Only one certain function.
Interpretation	clues linked to each proposition.	Shape, traces, material and comparison.	Similarity only.
Verification	Data capable of deciding between the leads.	Context, analysis, measurement, reference.	New guess.
Conclusion	Established, plausible and unknown distinguished.	The proposal remains revisable.	The doubt is erased.

Useful reminders

When an idea comes too quickly

"What exactly did you see?" Can you write the same sentence without naming the function? »

When the group gets stuck

"What other story would explain the same trace? What information would allow us to choose? »

What the correction does not contain

There is no “A = game, B = divination...” table: the objects have not yet been selected or validated. Even with a definitive corpus, certain functions may remain debated. The answer key should present the clues, sources and degree of confidence rather than a simplified answer.

Digital extension

The game “Coding with the knucklebone” can extend the discussion on contemporary technical use. It must be presented as a modern educational creation, not as the reconstruction of a documented ancient practice.

À VÉRIFIER

Complete this correction only after validation of the objects, notices, rights and interpretations by the scientific and museum managers.

BONE STORIES - ACTIVITY 4

Sources, credits and limits

Keep with the complete kit.

Method

This kit organizes a general educational approach to investigation: describe, question, propose several hypotheses, compare the context and conclude according to the degree of confidence. The precise archaeological formulations will depend on the future corpus.

Media excluded

The five Metropolitan Museum of Art tentative photographs associated with locations A through E in the redesign are not published in these PDFs. Their notices, their educational relevance and the chain of rights must still be validated.

digital game

“Coding with the ossicle” is a contemporary interactive extension. It constitutes neither an archaeological source, nor proof of ancient use, nor a validated reconstruction.

Status

Method templates for adult preparation and testing. Objects A-E, notices, sources, credits, interpretations and physical printing remain to be completed or validated.

Working translation: specialist scientific, educational and legal wording requires human review.

Traceability : **Activity page, updates and online credits.**